

CISCO SECURE WORKLOAD

FIPS 140-2/3 Compliance Framework

Cryptographic Module Enforcement — CSW-Relevant Scope

Prepared for: [Customer Name]

Prepared by: Cisco Systems, Inc.

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Regulatory Standard: FIPS 140-2 / FIPS 140-3 (CSW Enforcement Scope)

Environment: Hybrid (On-Premises + Cloud)

Deployment Stage: Fresh Installation

How to read the Coverage column

Direct means Secure Workload produces the primary evidence artifact for that row — segmentation, policy, flow and process telemetry, inventory, vulnerability and reachability, or forensic reconstruction. Direct is still evidence, not an attestation.

Supporting means Secure Workload contributes an input that feeds or complements another control, tool, or process — detection input only, reconciled against an external register, or one log source among several.

Evidence Required means Secure Workload has no record for that row. The customer supplies the contract, the analysis, or the HR / physical-security evidence from another control.

Every report in this set now uses the same three coverage words — Direct, Supporting, and Evidence Required. Coverage describes what evidence Secure Workload produces; it never means the control is passed.

Executive Summary & CSW Role

FIPS 140-2/3 is the US federal standard for cryptographic module validation. Cisco Secure Workload (CSW) is not itself a cryptographic module — it does not perform encryption. However, CSW plays a critical enforcement and detection role in FIPS compliance programs by detecting non-FIPS-compliant protocols, enforcing approved communication paths, auditing cryptographic service access, and defining the network boundary around FIPS-validated infrastructure.

This document is scoped exclusively to areas where CSW meaningfully contributes to FIPS compliance. Cryptographic module validation (CMVP), key management, and algorithm implementation remain outside CSW scope and require FIPS-validated hardware and software components.

CSW FIPS Contribution Summary

Role	Category	Protocol/ Area	CSW Implementation	Coverage
Detection	Non-FIPS Protocols	Plain HTTP (port 80)	Flow telemetry protocol filter identifies unencrypted HTTP traffic from sensitive workloads	Direct
Detection	Non-FIPS Protocols	Plain FTP (port 21)	Flow telemetry flags unencrypted file transfers from regulated environments	Direct
Detection	Non-FIPS Protocols	Telnet (port 23)	Flow telemetry detects legacy unencrypted remote access sessions	Direct
Detection	Non-FIPS Protocols	Plain LDAP (port 389)	Detects non-encrypted directory service access; enforces LDAPS migration	Direct
Detection	Non-FIPS Protocols	SNMPv1/v2 (port 161)	Identifies non-encrypted network management; enforces SNMPv3	Direct
Enforcement	Protocol Policy	Block Non-Compliant Protocols	CSW policy blocks all identified non-FIPS-compliant protocols at workload level	Direct
Enforcement	Access Control	Crypto Service Boundary	FIPS-Boundary scope restricts HSM/KMS access to approved workloads only	Direct
Audit	Telemetry	Crypto	Full flow audit trail for all connections to	Direct

		Service Access Log	FIPS-validated cryptographic services	
Audit	Process Monitoring	Crypto Library Inventory	Process monitoring identifies workloads using non-validated crypto libraries	Direct
Boundary	Scope Isolation	FIPS Module Boundary	CSW scope defines and enforces network boundary around FIPS-validated crypto infrastructure	Direct
Supplementary	Not in CSW Scope	Module Validation (CMVP)	Cryptographic module validation is performed by NIST-approved laboratories	Supplementary
Supplementary	Not in CSW Scope	Key Generation & Storage	Key lifecycle management requires FIPS-validated HSM hardware	Supplementary

Non-FIPS Protocol Detection & Enforcement

Detection via Flow Telemetry

CSW ADM and continuous flow telemetry identify non-FIPS-compliant protocols in real time across all monitored workloads:

Protocol	Port	FIPS Issue	CSW Detection	Approved Replacement	
HTTP	80	No encryption	Flow telemetry protocol filter	HTTPS (port 443)	Direct
FTP	21	No encryption	Flow telemetry port filter	SFTP/FTPS (port 22/990)	Direct
Telnet	23	No encryption	Flow telemetry port filter	SSH with FIPS ciphers (port 22)	Direct
Plain LDAP	389	No encryption	Flow telemetry port filter	LDAPS (port 636)	Direct
SNMPv1/v2	161	No encryption	Flow telemetry port filter	SNMPv3 (port 162)	Direct
TLS 1.0/1.1	443	Non-approved version	Process + flow analysis	TLS 1.2+ with FIPS cipher suites	Supporting

FIPS Boundary Architecture

CSW defines and enforces the network boundary around FIPS-validated cryptographic infrastructure:

- Create dedicated FIPS-Boundary scope containing HSM cluster, crypto services, and KMS endpoints
- Apply default-deny inbound policy — only explicitly approved workloads may access crypto services
- Log all connections to FIPS-Boundary scope with full process context for compliance audit
- Alert on any unapproved access attempt to FIPS-validated infrastructure

FIPS Compliance Evidence Package

Evidence Item	CSW Source	FIPS Relevance	Frequency
Non-FIPS protocol flow report	Investigate → Flow Search (port filter)	Protocol compliance	Weekly
FIPS-Boundary access log	Investigate → Flow Search (scope filter)	Module boundary	Continuous

Crypto service process audit	Investigate → Process Search	Module usage	Monthly
Protocol enforcement policy	Defend → Policy Analysis	Protocol enforcement	Monthly
HSM/KMS access audit	Flow Search → FIPS-Boundary scope	Key management audit	Continuous